CSM
Fall 2026

EE 64
Worksheet 8

1. Bode Plots for Filters

Bode plots provide us with a simple and easy tool to plot these transfer functions by hand. **Always remember that Bode plots are an approximation**; if you want the precisely correct plots, you need to use numerical methods (like solving using MATLAB or IPython).

When we make Bode plots, we plot the frequency and magintude on a logarithmic scale and the angle in either degrees or radians. We use the log scale because it allows us to break up complex transfer functions into its constituent components.

When plotting the transfer function, the most important quantity to look at is its cutoff frequency ω_c. We will take a look the individual Bode plots for low and high pass filters, and then look at how the Bode plot for a bandpass filter is constructed.

(a) Let's start by taking a look at $H(j\omega) = X(j\omega)Y(j\omega)$, the product of two transfer functions X and Y.

i. How can you graph $|H(j\omega)|$ using the graphs of $|X(j\omega)|$, $|Y(j\omega)|$?

ii. How does this compare when graphing $\log|H(j\omega)|$ using the graphs of $\log|X(j\omega)|$, $\log|Y(j\omega)|$?

iii. What is the relationship between $\angle H(j\omega), \angle X(j\omega)$, and $\angle Y(j\omega)$?

(b) Let $H(j\omega) = \frac{1}{1+j\frac{\omega}{\omega_c}}$. This transfer function models the effect of a low pass filter, and has cutoff frequency ω_c.

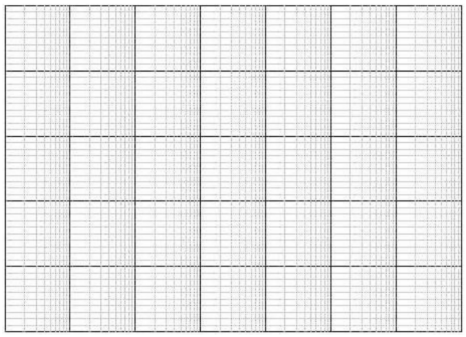
i. Find $|H(j\omega)|$ and $\angle H(j\omega)$.

ii. Again, **fill in the following table to analyze the magnitude and phase at the three regions $\omega << \omega_c$, $\omega = \omega_c$, and $\omega >> \omega_c$.**

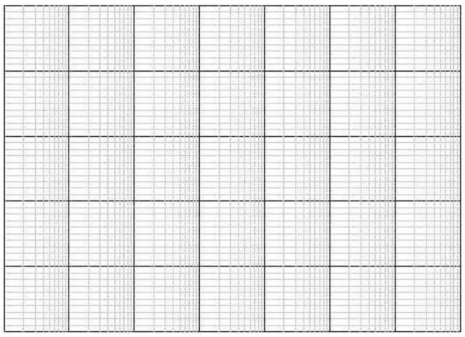
	$\omega << \omega_c$	$\omega = \omega_c$	$\omega >> \omega_c$
$H(j\omega)$			
$\lvert H(j\omega)\rvert$			
$\angle H(j\omega)$			

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iii. **Draw a sketch of $|H(j\omega)|$ and $\angle H(j\omega)$ using the points, filled in the chart above.**



$|H(j\omega)|$

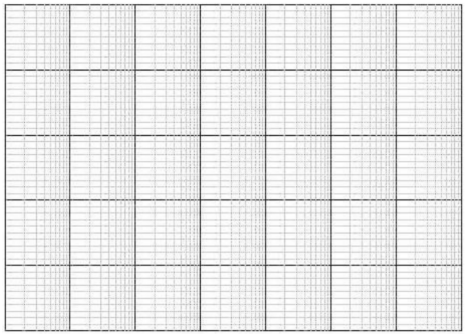


$\angle H(j\omega)$

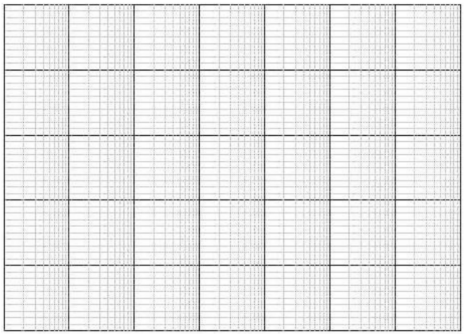
(c) Let $H(j\omega) = j\omega$. This transfer function does not have a cutoff frequency, but its behavior will prove to be useful when looking at a high-pass filter.

i. **Find $|H(j\omega)|$ and $\angle H(j\omega)$.**

ii. **Sketch them down below.**



$|H(j\omega)|$



$\angle H(j\omega)$

(d) Now let's go back to the low-pass filter with transfer function

$$H(j\omega) = \frac{1}{j\omega R_1 C_1 + 1} \tag{1}$$

Let $R_1 = 100\Omega$ and $C_1 = 100pF$

i. What is the cutoff frequency of this filter?

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ii. Sketch its phase and magnitude.

(e) Now consider the transfer function of a High-Pass Filter: $H(j\omega) = \frac{j\omega R_2 C_2}{1+j\omega R_2 C_2}$ where $R_2 = 1k\Omega$ and $C_1 = 10nF$

i. What is the cutoff frequency of this filter?

ii. Sketch its phase and magnitude.

(f) What happens if we cascade these two filters together with a unity-gain buffer in between them? Consider the resulting transfer function: $H(\omega) = \frac{1}{j\omega C_1 R_1+1} \cdot \frac{j\omega C_2 R_2}{j\omega C_2 R_2+1}$

i. What are the cutoff frequencies of this filter?

ii. Sketch its phase and magnitude. *Hint: How can we combine the plots of the individual filters together?*

Now lets take a look at two more concepts, **bandwidth** and **Q-factor**. The bandwidth of a resonance circuit is the difference between its upper and lower cutoff frequencies. It can also be expressed as *resonance frequency* divided by its *Q-factor*:

$$bandwidth = \Delta\omega = f_{high} - f_{low} = \frac{\omega_n}{Q} \tag{2}$$

The Q-factor, or quality factor, of a circuit is the ratio of power stored to power dissipated, but can also be more simply thought of as the measure of how good a circuit is, where a higher Q-factor is often more desirable. This can be expressed mathematically as:

$$Q = \frac{\omega_n}{\Delta\omega} \tag{3}$$

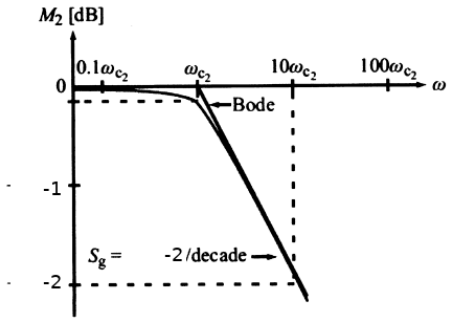
(g) Find the bandwidth of this circuit.

(h) What is its Q-factor? I.e.: $\frac{\omega_n}{\Delta\omega}$, where ω_n is the center frequency of the band.

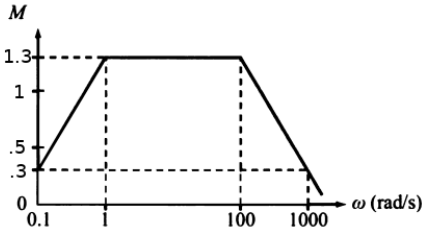
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2. Transfer Functions from Bode Plots

(a) **Derive a transfer function that would result in the following Bode plot.**



(b) **Derive a transfer function that would result in the following Bode plot.**



Hint: $\log_{10} 2 = 0.3$

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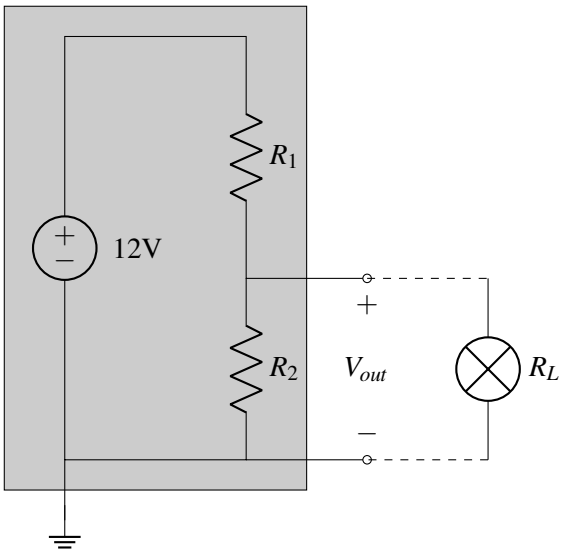
3. Op Amps as Buffers

Learning Goal: This problem helps understand the operating principle of an op-amp buffer and how it helps with loading.

Relevant Notes: *Note 19 Section 19.7* goes different op-amp circuit topology and corresponding derivations.

Now we will revisit a problem that you might have seen before, with our new knowledge of op-amps. We have access to a circuit inside a 'black box' as shown below, with two terminal coming out of it.

(a) We need a voltage of 6V power a light bulb with resistance R_L. Design R_1 and R_2 inside the black box so that the voltage across R_2 is exactly equal to this required voltage **when the bulb is not connected**; i.e. $V_{R_2} = V_{out} = 6\text{V}$.



(b) Now let us connect the bulb R_L across R_2. What is the voltage across R_1, R_2 and R_L when the bulb is connected when $R_L = R_2$? Use the values of R_1 and R_2 from the last part. Will the light bulb turn on? What happens if $R_L = 2R_2$?

(c) Using your knowledge of op-amps, how could you resolve this issue of V_{out} changing based on the value of R_L? Think about how you might use an op-amp buffer.

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